CSE 209 Computer Architecture: Past Question Analysis

Based on: 14 question papers (BUET CSE 209 from 2022-23 and 2023-24, plus CSE 305 from 2012-13 to 2022-23, which used the same textbook) and all 12 course slide sets (Ch 1, 2, 3, 4 Part 1 & 2, 5, ALU Design, MIPS Control Unit, GPUs).

Note on the numbers: marks are approximate. Some scanned pages were partly unreadable, and some sub-parts cover two topics. Use them to compare topics, not as exact figures.

Key findings

1. **Four topics account for roughly 60% of all marks:** MIPS assembly/ISA, Cache memory, Pipelining hazards and Single-cycle datapath. Each one appears in **every** paper.

2. **The CSE 209 papers (your course code) focus more on Chapters 4 and 5.** Pipelining, cache, virtual memory/TLB and datapath design together take about 60% of those two papers. Old I/O, disk, RAID and multiprocessor questions have disappeared and are not in your slides, so skip them.

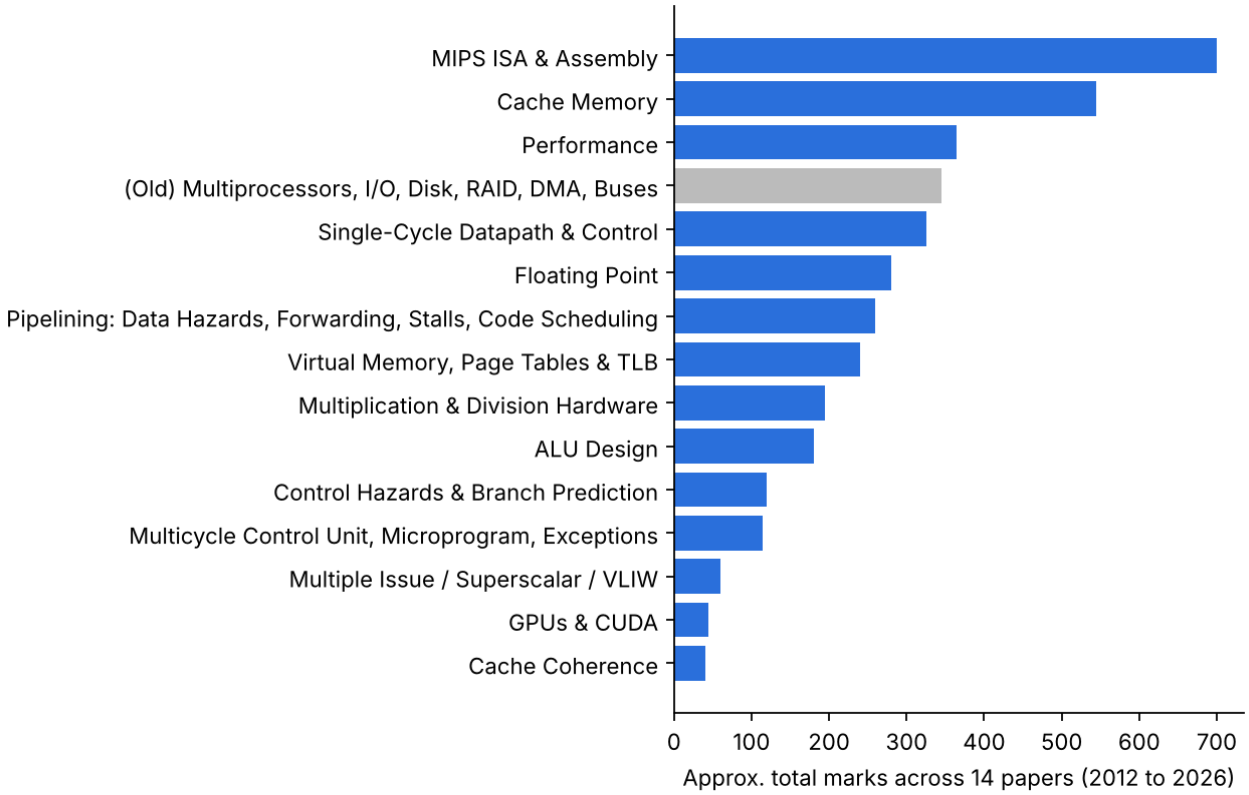
3. **Many questions repeat almost word for word**, such as the forwarding unit for double data hazards, the 1-bit predictor nested-loop problem, the 4 GHz / 200 ns multilevel cache problem, the three small caches (FA / 2-way / DM), the GPU+CUDA questions and the dual-issue loop. Section 5 lists them.

4. **The ALU input-equation design question (Xi, Yi, Zi) has appeared in all of the last 6 papers.** It is close to guaranteed marks (10 to 20) once you practise the method.

Section A vs Section B pattern (CSE 209)

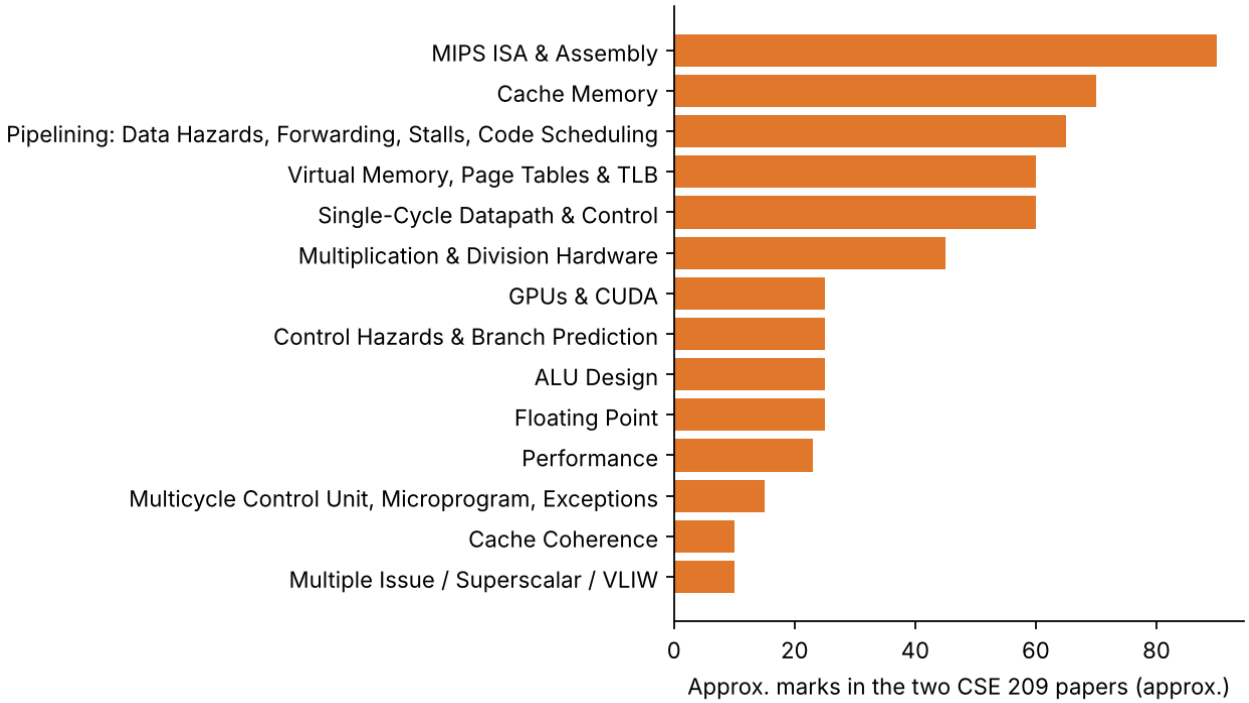
Section A (Q1 is mandatory): pipelining and hazards, cache, virtual memory/TLB, coherence, GPU and multiple issue (Ch 4 Part 2, Ch 5, GPU). **Section B** (Q5 is mandatory): datapath and instruction formats, ALU design, performance, MIPS coding, addressing modes, multiplication/division, IEEE 754 (Ch 1, 2, 3, ALU, Ch 4 Part 1). In both sections the mandatory question has been on **pipelining/forwarding** (A) and **datapath + ALU** (B).

1. Topic ranking: all 14 papers



Topic	Slides	~Marks (14 papers)	Papers it appears in	~Marks, 2 CSE 209 papers
MIPS ISA & Assembly (C to MIPS, procedures, encoding, addressing modes, branch range)	Ch 2	700	14 / 14	90
Cache Memory (mapping, traces, size/tag bits, multilevel CPI)	Ch 5	545	14 / 14	70
Pipelining: Data Hazards, Forwarding, Stalls, Code Scheduling	Ch 4 Pt 2	260	14 / 14	65
Performance (CPU time, CPI, Amdahl, great ideas, benchmarks)	Ch 1	365	13 / 14	23
Single-Cycle Datapath & Control	Ch 4 Pt 1	325	14 / 14	60
Floating Point (IEEE 754, bias, FP addition, FP code)	Ch 3	280	13 / 14	25
Virtual Memory, Page Tables & TLB	Ch 5	240	12 / 14	60
Multiplication & Division Hardware	Ch 3	195	13 / 14	45
ALU Design (input equations Xi,Yi,Zi, status bits)	ALU Design	180	9 / 14	25
Control Hazards & Branch Prediction	Ch 4 Pt 2	120	11 / 14	25
Multicycle Control Unit, Microprogram, Exceptions	Control Unit / Ch 4	115	6 / 14	15
Multiple Issue / Superscalar / VLIW	Ch 4 Pt 2	60	4 / 14	10
GPUs & CUDA	GPU slides	45	2 / 14	25
Cache Coherence (snooping)	Ch 5	40	5 / 14	10
Old-syllabus only: Multiprocessors, I/O, Disk, RAID, DMA, Buses	not in slides	345	10 / 14	0

2. What the CSE 209 papers emphasise



The two most recent papers are the best guide to your exam, because they were set for the same course code and slides. Pipelining, cache, VM/TLB and datapath design make up most of them. GPU/CUDA only appeared once (25 marks), but the questions were copied from 2021-22, so they are easy marks.

3. Recommended study order

#	Topic	Why	Slides
1	Pipelining: data hazards, forwarding, stalls, code scheduling (+ branch hazards)	Mandatory Q1 in Section A. Appears in all 14 papers. Repeats: forwarding unit with equations, cycle counting with stalls, A=B+C;D=A+E scheduling, flush and stall at the same time, branch prediction.	Ch 4 Part 2
2	Cache memory	Appears in all 14 papers and carries the most marks after ISA. Practise traces (DM / 2-way / FA with LRU), tag and total-bit calculations, multilevel cache CPI, write-through vs write-back.	Ch 5
3	Single-cycle datapath & control signals	Mandatory Q5 in Section B. Add bne/j/jr/jal to the datapath, set control signal values, explain why PC+4 and why the upper 4 bits are used in a jump.	Ch 4 Part 1
4	Virtual memory, page table, TLB	60 marks in the CSE 209 papers. TLB/page-table trace, page table size (single, multi-level, inverted), TLB/cache flowchart, which TLB/PT/cache combinations are possible.	Ch 5
5	ALU design	Appeared in all of the last 6 papers. Deriving Xi, Yi, Zi follows a fixed method, so marks are reliable.	ALU Design
6	MIPS assembly & ISA	The most marks overall, but spread over many small questions. Focus on recursive procedures, loops and if statements, large constants (lui/ori), instruction encoding, addressing modes, branch/jump range, and the floating-point code questions.	Ch 2
7	Arithmetic: multiplication/division hardware and IEEE 754	Draw the first and refined multiplier, trace a division step by step, find the bias, smallest/largest/denormalized values, and the FP addition steps.	Ch 3
8	Performance	Short numerical questions that are easy marks: CPI by instruction class, execution time, Amdahl's law, 'how much must CPI improve', eight great ideas.	Ch 1

#	Topic	Why	Slides
9	GPU/CUDA, multiple issue, cache coherence	Low frequency but repeated word for word. Learn the model answers for: PC-GPU connection, kernel/grid/block mapped to SP/SM, dual-issue scheduling + IPC + loop unrolling, snooping.	GPU, Ch 4 Pt 2, Ch 5
10	Multicycle control unit / exceptions	Occasional. Know the EPC and Cause registers, precise exceptions in the pipeline, and the FSM/microprogram basics.	Control Unit

4. Topic-by-topic breakdown with typical questions

Pipelining (Ch 4 Part 2)

• Design a full forwarding unit with equations, considering double data hazards (2014, 2015, 2017, 2019, 2021, 2024, 2026)

• Count total cycles/stalls for lw/add/sw sequences, with and without forwarding, then reschedule the code (nearly every year)

• Hazard detection unit for load-use: conditions and signals (2025)

• Data hazards for branches: examples needing 0, 1 and 2 stalls (2014, 2017, 2023)

• 1-bit predictor problem with nested loops, and the 2-bit predictor FSM (5 papers)

• 'Even with a perfect predictor, a taken branch costs 1 cycle': explain why (2015, 2017, 2018, 2026)

• Can a flush and a stall happen at the same time? (2019, 2025). Jump stalls and moving control to IF (2025)

• Hazards with branches resolved in ID: total cycles for taken and not-taken cases (2026)

Cache memory (Ch 5)

• Three small caches (FA, 2-way, DM; 4 or 8 blocks; LRU): count misses (8+ papers)

• Direct-mapped trace with byte addresses 0, 4, 16, 132, 232, ...: block size, entries, replacements, hit ratio (2016, 2026)

• Total bits / tag bits for an n-way cache (very frequent)

• Multilevel cache: base CPI 1, 4 GHz, 200 ns, add L2 then L3 (2017, 2021, 2026). Compare P1 and P2 (2025)

• I-cache/D-cache miss rates and how much faster a perfect cache would be

• Write-through vs write-back, write buffer, larger blocks, early restart, critical word first, locality

Virtual memory & TLB (Ch 5)

• TLB + page table trace with 4 KB pages and LRU, and the final state (2026)

• Page table size: single-level, 3-level, inverted (2019, 2025)

• Flowchart of the TLB and data cache for read/write (2015, 2017, 2026)

• Which TLB / page table / cache hit-miss combinations are possible (2021)

• Handling a TLB miss, flushing on context switch / ASID (2019, 2020, 2025)

Single-cycle datapath (Ch 4 Part 1)

• Add bne, j, jr, jal, slt, lui to the datapath (2013, 2019, 2025, 2026)

• Control signal values for add, sw, beq (2015, 2020)

• Draw the datapath for {add, lw, beq} or {add, sw, beq}

• Single-cycle vs pipeline speedup for 3 lw instructions plus N more

ALU design

• Derive Xi, Yi, Zi for a table of arithmetic and logic functions (every paper since 2018)

• Status bits V, Z, S, C. Build a comparator for A>B, A=B, A<B (2020, 2021)

• Why is the overflow bit the XOR of the last two carries (2025)

MIPS ISA & assembly (Ch 2)

• Recursive C to MIPS: fun(a,b), fib, sum(n), rstrcopy, sort/swap

• Loops and if statements, a constant like 131075 needing lui/ori, float/double code

• Encode instructions into machine code fields, bne/j immediate values

• Reachability: can a j/beq instruction reach a given address? Fixing an out-of-range beq

• Addressing modes (asked in 10+ papers), instruction formats

• Design principles, synchronization (ll/sc), static vs dynamic linking, JIT, pseudo-instructions

Arithmetic (Ch 3)

• Multiplication hardware (first vs refined) with flowchart and a 4-bit trace

• Division step-by-step with register values

• IEEE 754: bias, smallest/largest/denormalized values for custom bit widths, decode a hex value

• FP addition flowchart, overflow/underflow, guard/round/sticky bits

Performance (Ch 1)

• CPI by instruction class, comparing P1 and P2, 'how much must CPI of class X improve'

• Clock rate target (10 s at 4 GHz, 1.2x cycles)

• Amdahl's law, eight great ideas, SPEC geometric mean

GPU & multiple issue

• PC + GPU connection diagram for Intel/AMD, and how CPU and GPU access each other's memory (2023, 2026)

• CUDA kernel/grid/thread block mapped to SP/SM (2023, 2026)

• Dual-issue scheduling of the lw/addu/sw/addi/bne loop, IPC, loop unrolling (2023, 2026)

5. Repeated questions to practise first

These have been asked two or more times with almost the same wording. Solving each one once covers a large share of the marks.

Question	Years seen	Marks
Forwarding unit + equations (double data hazard)	2014, 2015, 2017, 2019, 2021, 2024, 2026	10-15
A=B+C; D=A+E cycles and code scheduling	2015, 2017, 2018, 2020, 2023, 2025	10-15
lw/add/sw sequence: hazards, stalls, rescheduling	2015, 2016, 2017, 2020, 2023	10-15
1-bit predictor nested loop	2014, 2015, 2017, 2023, 2026	5-10
Perfect predictor still costs 1 cycle	2015, 2017, 2018, 2026	5-10
Three small caches FA/2-way/DM	2013, 2015, 2016, 2017, 2018, 2020, 2023	15
Multilevel cache, 4 GHz, 200 ns, L2 then L3	2016, 2017, 2023, 2026	10
Direct-mapped byte reference trace	2016, 2026	10-15
TLB-cache read/write flowchart	2016, 2017, 2018, 2026	5-10
Page table works with VA/PA/memory/disk	2016, 2017, 2023	5-10
Snooping protocol example	2023, 2026	10
GPU connection + CUDA mapping	2023, 2026	25
Dual-issue loop + IPC	2023, 2026	10-15
ALU Xi/Yi/Zi design	2018, 2019, 2020, 2021, 2023, 2024, 2025, 2026	10-20
Addressing modes illustration	2013, 2014, 2015, 2016, 2019, 2021, 2023, 2025, 2026	9-15
beq out of range / jump reachability	2013, 2014, 2015, 2016, 2019, 2021, 2025, 2026	5-10
if (x > y+131075) in MIPS	2020, 2024	8
Float/double code: if(a+b>c) x=(y*z)/w	2021, 2024	7-12
Multiplication hardware first/refined	2014, 2015, 2016, 2019, 2020, 2025, 2026	12-20
Division trace with a 4/6-bit ALU	2018, 2019, 2025	10-15
Instruction-class CPI and 'improve CPI by'	2015, 2016, 2017, 2018, 2019, 2023, 2024	15-20
Eight great ideas / MIPS design principles	2015, 2016, 2017, 2021, 2023	10-15
MIPS synchronization (ll/sc)	2014, 2015, 2016, 2017, 2024	5-10
DLL / JIT / static vs dynamic linking	2021, 2024, 2025	5-15

6. A practical plan

Week 1: Ch 4 Part 2 (pipelining) and Ch 5 (cache + VM). Solve every repeated question above for those chapters. **Week 2:** Ch 4 Part 1 datapath, ALU design, then Ch 2 MIPS coding (one recursive function every day). **Week 3:** Ch 3 arithmetic, Ch 1 performance numericals, GPU/multiple-issue model answers, then a full timed mock using the 2023-24 paper.

Skip or only skim: I/O, DMA, interrupts, disk and RAID, bus design, the single-bus CPU, interconnection networks, multiprocessor C code. They only appear in old CSE 305 papers and are not in your slides.